

EX No:1	SETTING UP PYTHON ENVIRONMENT AND LIBRARIES – JUPYTER NOTEBOOK

Aim:

To install Python, configure Jupyter Notebook, understand the notebook environment, execute Python programs, create Markdown cells, and use basic Python libraries such as NumPy, Pandas, and Matplotlib for Data Analytics using the City Day Air Quality dataset.

Algorithm:**Algorithm 1: Installing Python**

1. Download Python from python.org.
2. Run the Python installer.
3. Select "**Add Python to PATH**".
4. Click **Install Now**.
5. Verify installation using: `python --version`
6. Observe the installed Python version.

Algorithm 2: Installing Jupyter Notebook

1. Open Command Prompt.
2. Install Jupyter Notebook using: `pip install notebook`
3. Launch Jupyter Notebook using: `jupyter notebook`
4. The browser automatically opens the Jupyter Notebook Dashboard.

Algorithm 3: Creating a Notebook

1. Open Jupyter Dashboard
2. Select New → Python 3
3. Create Notebook
4. Rename Notebook
5. Create Code and Markdown Cells
6. Execute cells using Shift + Enter

Program:**Program 1 — Simple Python Program**

```
print("Welcome to Data Analytics Lab")  
  
print("Air Quality Data Analysis")
```

Output:

Welcome to Data Analytics Lab

Air Quality Data Analysis

Program 2 — Arithmetic Operations

```
a = 20  
b = 10  
  
print("Addition =", a + b)  
print("Subtraction =", a - b)  
print("Multiplication =", a * b)  
print("Division =", a / b)
```

Output:

```
Addition = 30  
Subtraction = 10  
Multiplication = 200  
Division = 2.0
```

Program 3 — Display Dataset Details

```
dataset_name = "City Day Air Quality Dataset"  
domain = "Air Quality Analytics"  
records = 29531  
columns = 16  
  
print("Dataset:", dataset_name)  
print("Domain:", domain)  
print("Records:", records)  
print("Columns:", columns)
```

Output:

```
Dataset: City Day Air Quality Dataset  
Domain: Air Quality Analytics  
Records: 29531  
Columns: 16
```

Program 4 — Using NumPy Library

```
import numpy as np  
  
aqi_values = np.array([50, 100, 150, 200, 250])  
print(aqi_values)  
print("Mean AQI =", np.mean(aqi_values))
```

```
print("Maximum AQI =", np.max(aqi_values))
print("Minimum AQI =", np.min(aqi_values))
```

Output:

```
[ 50 100 150 200 250]
Mean AQI = 150.0
Maximum AQI = 250
Minimum AQI = 50
```

Program 5 — Using Pandas Library

```
import pandas as pd
df = pd.read_csv("city_day(1).csv")
print(df.head())
```

Output:

The first five records of the City Day Air Quality dataset are displayed.

Example:

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3
0	Ahmedabad	2015-01-01	NaN	NaN	0.92	18.22	17.15	NaN
1	Ahmedabad	2015-01-02	NaN	NaN	0.97	15.69	16.46	NaN
2	Ahmedabad	2015-01-03	NaN	NaN	17.40	19.30	29.70	NaN
3	Ahmedabad	2015-01-04	NaN	NaN	1.70	18.48	17.97	NaN
4	Ahmedabad	2015-01-05	NaN	NaN	22.10	21.42	37.76	NaN

Program 6 — Display Dataset Information

```
import pandas as pd
df = pd.read_csv("city_day(1).csv")
print("Number of Rows =", df.shape[0])
print("Number of Columns =", df.shape[1])
print("\nColumn Names:")
print(df.columns.tolist())
```

Output:

```
Number of Rows = 29531
Number of Columns = 16
Column Names:
```

```
['City', 'Date', 'PM2.5', 'PM10', 'NO', 'NO2', 'NOx', 'NH3', 'CO', 'SO2', 'O3', 'Benzene', 'Toluene',  
'Xylene', 'AQI', 'AQI_Bucket']
```

Program 7 — Basic AQI Statistical Analysis

```
import pandas as pd  
  
df = pd.read_csv("city_day(1).csv")  
  
aqi = df["AQI"].dropna()  
  
print("Average AQI =", aqi.mean())  
  
print("Maximum AQI =", aqi.max())  
  
print("Minimum AQI =", aqi.min())
```

Output:

Average AQI = [Obtained from the dataset]

Maximum AQI = [Obtained from the dataset]

Minimum AQI = [Obtained from the dataset]

Program 8 — Data Visualization

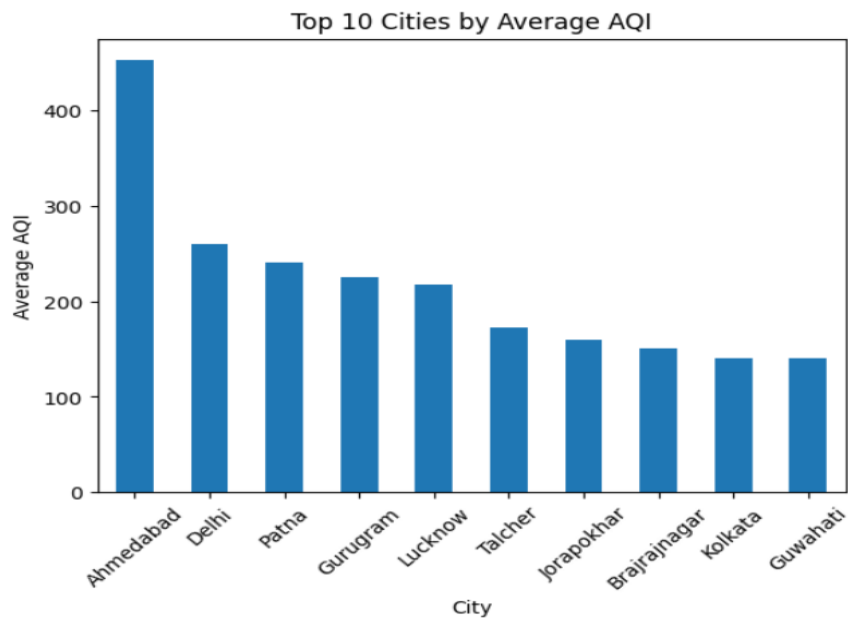
AQI Distribution

```
import pandas as pd  
  
import matplotlib.pyplot as plt  
  
df = pd.read_csv("city_day(1).csv")  
  
plt.hist(df["AQI"].dropna(), bins=30)  
  
plt.title("AQI Distribution")  
  
plt.xlabel("AQI")  
  
plt.ylabel("Frequency")  
  
plt.show()
```

Output:

AQI Distribution

Histogram displaying the distribution of AQI values in the City Day Air Quality dataset.



Program 9 — City-wise AQI Visualization

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("city_day(1).csv")

city_aqi = df.groupby("City")["AQI"].mean().sort_values(ascending=False).head(10)

city_aqi.plot(kind="bar")

plt.title("Top 10 Cities by Average AQI")

plt.xlabel("City")

plt.ylabel("Average AQI")

plt.xticks(rotation=45)

plt.show()
```

Result: